1. Environment Setup

Installing Python

You can check if Python is already installed on your system, and whether its version is up to date. Open the Terminal (MacOS) or Command Prompt (Windows) and type:

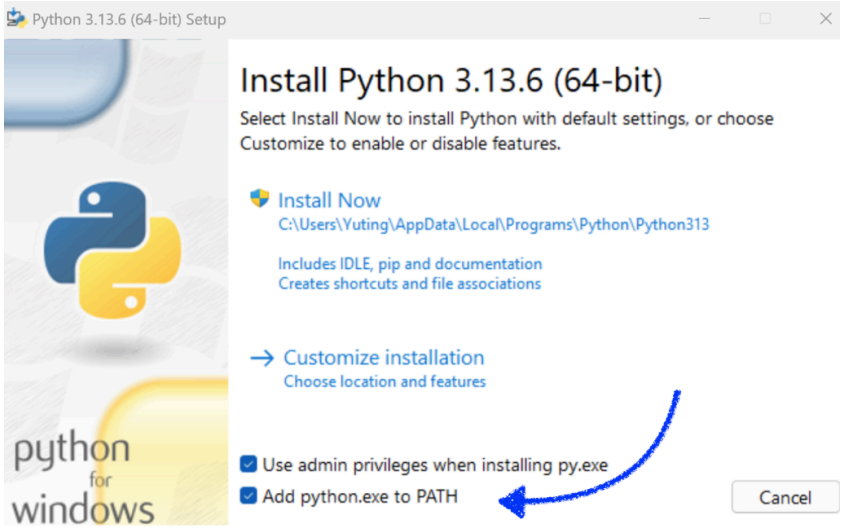
```
python3 --version
```

Please ensure that you are using at least **Python 3.12**.

If you don't have Python installed or if the version is lower than 3.12, follow the steps below.

Go to the official Python download page to obtain the appropriate installer for your operating system.

- The installation for Mac users is straightforward.
- **For Windows users,** ensure to check the boxes for **Use admin privileges** and **Add python.exe to PATH** during installation.



After the installation is complete, close and re-open your Terminal/Command Prompt and run the same command again.

```
python3 --version
```

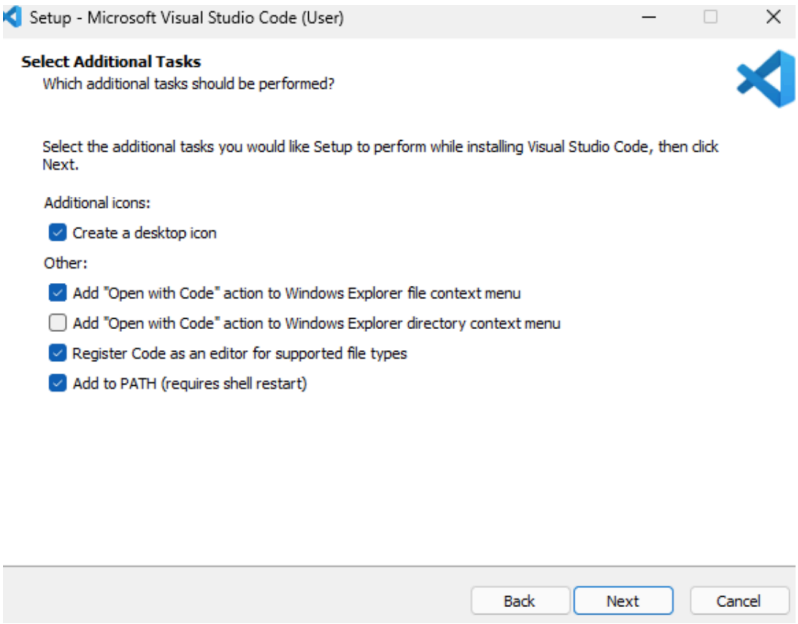
If Python is installed properly, it will show the version number.

Installing VS Code and Extensions

To run the notebook, we will use **Visual Studio Code (VS Code)** as our code editor.

Go to the official website and obtain the appropriate installer for your operating system.

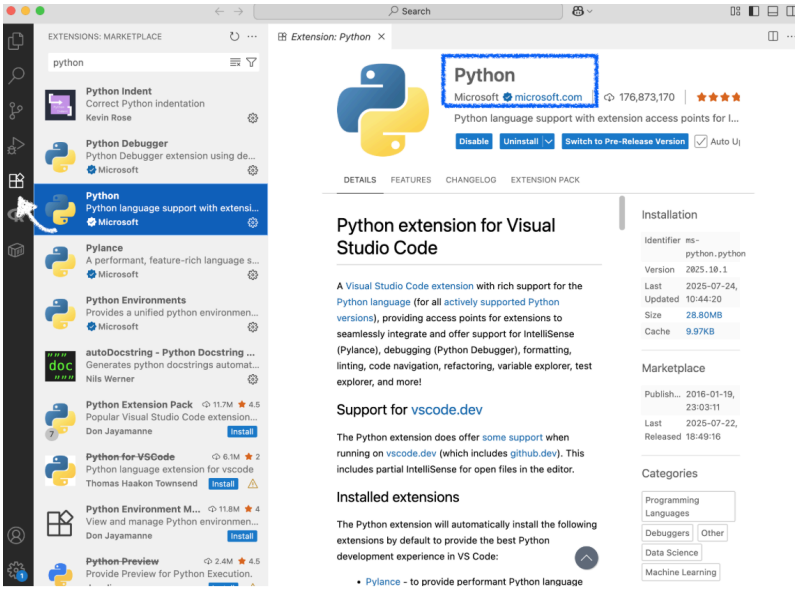
On Windows, also make sure to check **Register Code as an editor** and **Add VS Code to PATH**.



After installation is completed, launch VS Code.

Then, navigate to the **Extension** tab (the squares button on the left sidebar) and install

- The **Python** extension by Microsoft.
- **Jupyter** and **Jupyter Notebook Renderer** extensions for notebook support.



We will need to **restart VS Code** after installing Python and the VS Code extensions.

Installing packages

Once the environment is active, we can install packages with `pip install`.

The file **requirements.txt** contains a set of packages required to run the notebook. Download and save it in the same folder as the .ipynb file. Then run the following command:

```
pip install -r requirements.txt
```

Checking the setup

To check the current environment setup, the installed packages, and their dependencies:

```
pip list
```

2. Instructions to Run the Notebook

1. Ensure the ‘dataset’ folder is in the same directory as the ‘project_IT1244.ipynb’ file.
2. Open the notebook in VS Code and run the cells in numerical order to prevent any errors.
3. For section ‘2.1 Cropping’, uncomment the last line and run it once only.
   `#stats = crop()`
4. Similarly, for section ‘6. Labelling’, run the code cell only once otherwise an error will occur.